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In [2]: #CIA2 - 2460601 (A Deepak)
#Load Dataset
import pandas as pd
import numpy as np
df=pd.read_csv('paper_moisture_control.csv')
df
```

Out[2]:

	timestamp	machine_id	batch_id	paper_grade	moisture_pct	temperature_C	humidity_pct	cycle_start	cycle_end	deviation_flag
0	2025-09-01 07:00:00	M004	BATCH_001	coated	6.229510	28.048834	56.041926	2025-09-01 07:00:00	2025-09-01 09:00:00	0
1	2025-09-01 07:10:00	M004	BATCH_001	coated	6.525639	24.146738	54.039701	2025-09-01 07:00:00	2025-09-01 09:00:00	0
2	2025-09-01 07:20:00	M004	BATCH_001	coated	6.528640	22.919657	54.670740	2025-09-01 07:00:00	2025-09-01 09:00:00	0
3	2025-09-01 07:30:00	M004	BATCH_001	coated	6.439983	29.414033	33.000346	2025-09-01 07:00:00	2025-09-01 09:00:00	0
4	2025-09-01 07:40:00	M004	BATCH_001	coated	7.116764	23.824238	45.115315	2025-09-01 07:00:00	2025-09-01 09:00:00	0
...	...	...	...	...	...	...	...	...	...	...
1295	2025-09-17 20:20:00	M001	BATCH_100	coated	6.822513	23.943807	46.186320	2025-09-17 19:00:00	2025-09-17 21:00:00	0
1296	2025-09-17 20:30:00	M001	BATCH_100	coated	6.698634	23.938234	NaN	2025-09-17 19:00:00	2025-09-17 21:00:00	0
1297	2025-09-17 20:40:00	M001	BATCH_100	NaN	6.695514	26.023287	48.055718	2025-09-17 19:00:00	2025-09-17 21:00:00	0
1298	2025-09-17 20:50:00	M001	BATCH_100	coated	5.969523	27.106910	48.089398	2025-09-17 19:00:00	2025-09-17 21:00:00	0
1299	2025-09-17 21:00:00	M001	BATCH_100	coated	6.632171	23.921920	46.832578	2025-09-17 19:00:00	2025-09-17 21:00:00	0

1300 rows × 10 columns

```
In [4]: #Convert timestamps to datetime objects
time_cols = ['timestamp', 'cycle_start', 'cycle_end']
for col in time_cols:
    df[col] = pd.to_datetime(df[col]) #This line must be indented
print(df[time_cols].dtypes)
```

```
timestamp      datetime64[ns]
cycle_start    datetime64[ns]
cycle_end      datetime64[ns]
dtype: object
```

```
In [4]: #NumPy-based stats
moisture_arr = df['moisture_pct'].to_numpy()
avg_moisture = np.mean(moisture_arr)
var_moisture = np.var(moisture_arr)
print(f"Dataset Shape: {df.shape}")
print(f"Average Moisture: {avg_moisture:.4f}%")
print(f"Moisture Variance: {var_moisture:.4f}")
```

```
Dataset Shape: (1300, 10)
Average Moisture: nan%
Moisture Variance: nan
```

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In [7]: #(Adding a small noise factor to a copy of data to simulate sensor jitter)
np.random.seed(42)
synthetic_noise = np.random.normal(0, 0.05, size=len(df))
df['moisture_synthetic'] = df['moisture_pct'] + synthetic_noise
df['moisture_synthetic']
```

```
Out[7]: 0      6.254346
1      6.518725
2      6.561024
3      6.516134
4      7.105057
...
1295   6.817568
1296   6.744588
1297   6.681000
1298   5.982892
1299   6.648256
Name: moisture_synthetic, Length: 1300, dtype: float64
```

```
In [14]: #2.Timestamp Alignment
#Aligning sensor readings
df['is_aligned'] = (df['timestamp'] >= df['cycle_start']) & (df['timestamp'] <= df['cycle_end'])
misaligned_count = (~df['is_aligned']).sum()
print(f"Rows with timestamp misalignment: {misaligned_count}")
```

Rows with timestamp misalignment: 0

```
In [10]: df_clean = df[df['is_aligned']].copy()  
df_clean
```

Out[10]:

	timestamp	machine_id	batch_id	paper_grade	moisture_pct	temperature_C	humidity_pct	cycle_start	cycle_end	deviation_flag	moisture_synthetic	is_aligned
0	2025-09-01 07:00:00	M004	BATCH_001	coated	6.229510	28.048834	56.041926	2025-09-01 07:00:00	2025-09-01 09:00:00	0	6.254346	True
1	2025-09-01 07:10:00	M004	BATCH_001	coated	6.525639	24.146738	54.039701	2025-09-01 07:00:00	2025-09-01 09:00:00	0	6.518725	True
2	2025-09-01 07:20:00	M004	BATCH_001	coated	6.528640	22.919657	54.670740	2025-09-01 07:00:00	2025-09-01 09:00:00	0	6.561024	True
3	2025-09-01 07:30:00	M004	BATCH_001	coated	6.439983	29.414033	33.000346	2025-09-01 07:00:00	2025-09-01 09:00:00	0	6.516134	True
4	2025-09-01 07:40:00	M004	BATCH_001	coated	7.116764	23.824238	45.115315	2025-09-01 07:00:00	2025-09-01 09:00:00	0	7.105057	True
...	...	...	...	...	...	...	...	...	...	...	...	...
1295	2025-09-17 20:20:00	M001	BATCH_100	coated	6.822513	23.943807	46.186320	2025-09-17 19:00:00	2025-09-17 21:00:00	0	6.817568	True
1296	2025-09-17 20:30:00	M001	BATCH_100	coated	6.698634	23.938234	NaN	2025-09-17 19:00:00	2025-09-17 21:00:00	0	6.744588	True
1297	2025-09-17 20:40:00	M001	BATCH_100	NaN	6.695514	26.023287	48.055718	2025-09-17 19:00:00	2025-09-17 21:00:00	0	6.681000	True
1298	2025-09-17 20:50:00	M001	BATCH_100	coated	5.969523	27.106910	48.089398	2025-09-17 19:00:00	2025-09-17 21:00:00	0	5.982892	True
1299	2025-09-17 21:00:00	M001	BATCH_100	coated	6.632171	23.921920	46.832578	2025-09-17 19:00:00	2025-09-17 21:00:00	0	6.648256	True

1300 rows × 12 columns

```
In [16]: # 3. Data Cleansing and Imputation
initial_count = len(df_clean)
df_clean.drop_duplicates(inplace=True)
print(f"Duplicates removed: {initial_count - len(df_clean)}")
```

Duplicates removed: 0

```
In [18]: #Impute missing moisture readings (Forward-fill and Interpolation)
df_clean.loc[df_clean.sample(frac=0.01, random_state=42).index, 'moisture_pct'] = np.nan
df_clean['moisture_pct'] = df_clean['moisture_pct'].ffill().interpolate(method='linear')
df_clean['moisture_pct']
```

```
Out[18]: 0      6.229510
1      6.525639
2      6.528640
3      6.439983
4      7.116764
...
1295    6.822513
1296    6.698634
1297    6.695514
1298    5.969523
1299    6.632171
Name: moisture_pct, Length: 1300, dtype: float64
```

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In [ ]:
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